Long-context Reasoning for Agentic LLMs

Speakers

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Problem Statement

- **Motivation:** LLM agents rely on static prompts and Supervised Fine-Tuning (SFT), which struggle with long-horizon tasks
- Agentic LLMs reframes the problem as a policy over tools, memory, and language.
- **Key Paradigm:** Transition from SFT to RL with Verifiable Rewards (RLVR), where rewards are defined on interpretable outcomes (tool correctness, solution validity, consistency of reasoning traces)
- **Self-Evolving Agents:** Frameworks like Tool-R0 and Tool Zero show that self-play RL and pure RL from scratch can induce complex tool-calling behavior without curated human trajectories.
- **Our Focus:** Use RL to jointly optimize (1) tool selection, (2) action execution, and (3) graph-based retrieval for better pretraining and inference-time self-improvement across multi-session, long-context reasoning.

References

1. https://arxiv.org/abs/2602.05665
2. https://arxiv.org/html/2602.21320v1

02

RL for Reasoning

Problem: Outcome-only rewards (correct/incorrect answer) create gradient conflict.
For example: Good intermediate tool calls are penalized when the final answer is wrong, which slows down learning

Our Approach: Decompose complex workflows into a high-level Router that proposes subgoals and tool plans, and a low-level Solver that executes them with concrete calls and reasoning.

Benefit: Enables agents to handle complex, multi-turn tasks while learning efficiently and avoiding conflicts between planning and execution.

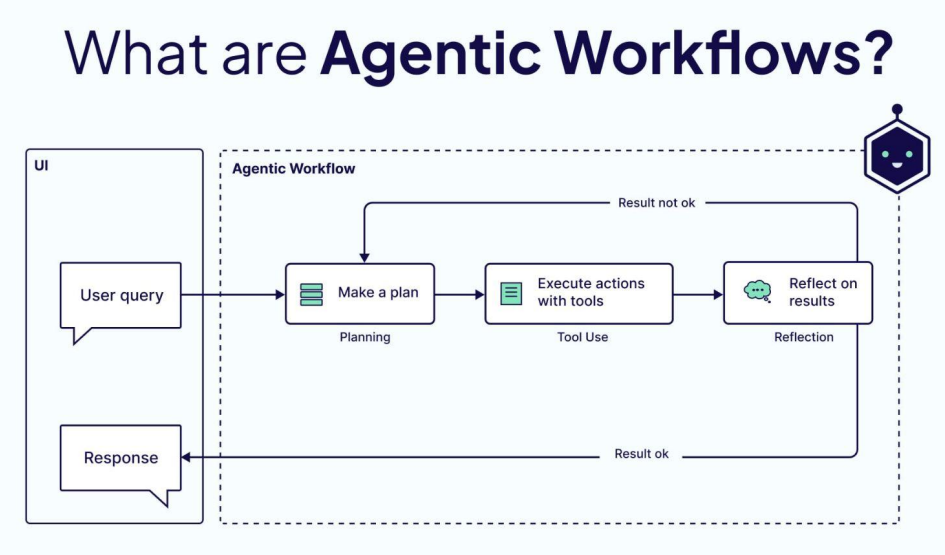
References

1. https://arxiv.org/html/2602.21320v1
2. https://ui.adsabs.harvard.edu/abs/2025arXiv250411536F/abstract

03

The Intractability of Multi-turn Workflows

- The utility of Agentic LLMs scales with the complexity of the workflows they can execute
- Multi-turn workflow complexity scales nonlinearly with number of tasks [1]
- Self-improvement provides inference time optimization for task-specific context; prioritizing local environment alignment over generalized capabilities
- Directed self-improvement targets workflow-specific failure modes, compounding gains over repeated interaction



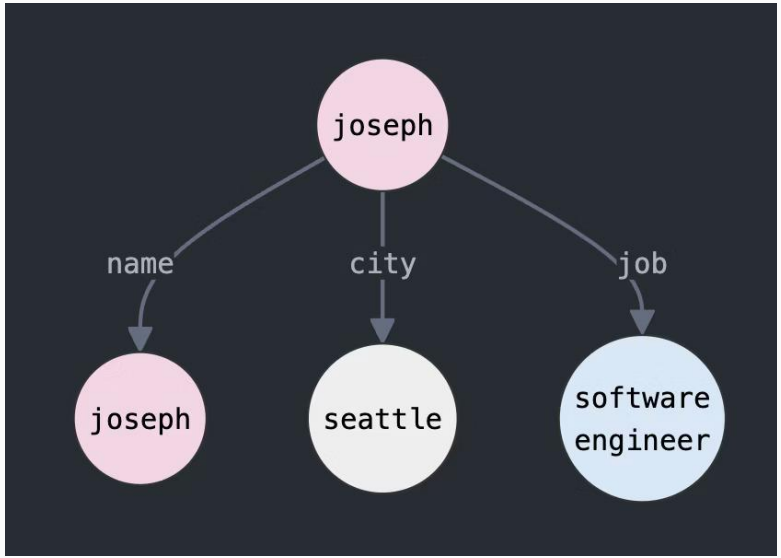
References

1. https://arxiv.org/pdf/2203.11171

04

Graph-based Memory

- Memory models provide an inference time adaptive knowledge store matching the inductive bias of adaptive self-improvement
- Graph based memory provide a structured memory primitive for storing more faithful representations of multi-turn (state, action) sequences
- We aim to improve the integrity of structured reflection workflows by combining external graph structures with internal working memory
- The KG-Agent framework offers a concrete template for integrating external knowledge graphs with internal knowledge memory



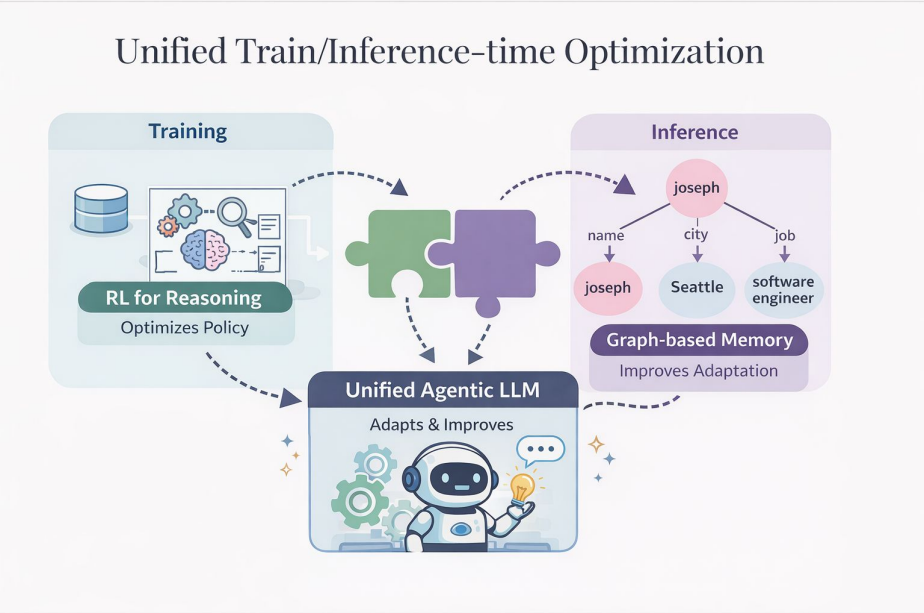
References

1. https://arxiv.org/pdf/2402.11163

05

Unified Train/Inference-time Optimization

- RL for reasoning optimizes the agent's general decision-making policy at training time
- Graph-based memory enables inference-time self-improvement through structured retrieval and reflection
- Jointly, they align long-horizon learning with context-specific adaptation
- This creates a unified agent architecture that improves both across tasks and within repeated workflows



Thank You